

BHAVIN BALDOTA

AI/ML Research Engineer — Generative AI, LLMs & Agentic Systems

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EDUCATION

Vishwakarma University, Pune, India

2021 – 2025

Bachelor of Technology (B.Tech) in Artificial Intelligence and Data Science

- **CGPA: 9.01 / 10** — Ranked 2nd in department (cohort of AI & Data Science engineers)
- **Perfect 10.0/10 CGPA** in both Semester 7 and Semester 8, achieved while working at DRDO and Full Time as AI/ML Engineer at Provilac respectively.
- Relevant Coursework: Deep Learning, Natural Language Processing, Computer Vision, Reinforcement Learning, Data Structures & Algorithms, Cloud Computing

RESEARCH EXPERIENCE

Research & Development Intern, Vishwakarma University

2022 – 2024

Supervisor: Prof. Kailas Patil, PhD (National University of Singapore), Dean – School of Science and Technology

- Built and evaluated deep learning models for disease and defect detection across five domains — coconut tree pathology, sugarcane leaf disease, lung cancer, brain tumour, and pothole detection (abrasion & cavitation) — each achieving 90%+ accuracy using TensorFlow and Python
- Contributed to two of these datasets being published in Elsevier's Data in Brief, now used by the wider machine-learning research community

Research Intern (AI/ML), DRDO

Jul 2024 – Dec 2024

Defence Research and Development Organisation, India

- Developed deep learning models for AI-based audio perception and sound source localization, achieving 92% localization accuracy across multi-channel acoustic environments for a defence application

Independent Research (in progress)

2025 – 2026

- Writing 2–3 research papers as first author on applied deep learning and agentic-AI system reliability, including a paper on the DeliverIQ system (see Selected Projects) targeted at IEEE Access
- Core research interest: reliability and evaluation of LLM-based agentic systems — specifically, how compounding errors in multi-step tool-calling pipelines can be detected and corrected, and how to build evaluation frameworks for agents operating in open-ended, multi-step environments

RESEARCH CONTRIBUTIONS TO PUBLISHED WORK

- **Coconut (Cocos nucifera) Tree Disease Dataset.** Elsevier, Data in Brief. Contributed to ML model development, training pipeline, and evaluation for plant disease classification.
- **Natural Pothole Dataset (Abrasion & Cavitation Effects).** Elsevier, Data in Brief. Contributed to CV-based road damage detection model training and dataset curation.

PROFESSIONAL EXPERIENCE

Lead Software Engineer – GenAI, Persistent Systems

Sep 2025 – Present

Pune, India

- Architected 5+ enterprise GenAI systems — RAG pipelines and AI assistants handling 10,000+ daily queries at sub-200ms latency
- Engineered production AI microservices (FastAPI + ChromaDB/FAISS) for semantic search across 50,000+ documents at 99.5% uptime
- Reduced LLM hallucination by 35% via prompt engineering and custom evaluation pipelines; cut inference cost by 25% through model optimization
- Designed LangGraph multi-agent orchestration with tool-calling and reflection loops, cutting manual workflow time by 60%
- Mentor 3 junior engineers; lead cross-functional delivery of production AI solutions with product, data, and platform teams

AI & ML Engineer, Provilac (P9 Farms Pvt Ltd)

Nov 2024 – Aug 2025

Pune, India

- Independently designed, developed, and deployed 14 production AI/ML systems spanning SaaS backend architecture, logistics intelligence, computer vision, and generative AI/LLM systems, including:
- a multi-tool agentic system (LangGraph) with 6 callable business-operation tools and context-aware API execution; a RAG-based company knowledge assistant (LangChain, FAISS, ChromaDB) with hallucination reduction through grounding; a full-stack LLM customer-support chatbot (FastAPI/Flask); a GPS-based delivery-proximity verification system with dynamic confidence scoring; a Text-to-SQL system for natural-language database queries; and a KYC verification bot combining OCR with LLM-based semantic validation
- Improved process automation efficiency by 30% through combined deep learning, computer vision, and GenAI deployments

Software Developer, Provilac (P9 Farms Pvt Ltd)

Jun 2020 – 2023

Pune, India

- Built and maintained customer-facing and logistics applications serving 10,000+ active users, including subscription-based ordering, delivery scheduling, and real-time tracking
- Built internal route-optimization and delivery-management tools, reducing operational errors by 40%
- Improved backend API performance via FastAPI and Firebase integration, cutting application load times by ~50% and improving customer retention by 30%

Expert Collaborator, Innovation Hub – MIT-WPU

Apr 2025 – Dec 2025

Dr. Vishwanath Karad MIT World Peace University

- Invited under a formal Memorandum of Understanding to run workshops on Generative AI and LLMs, mentor student teams for hackathons, and co-develop research proposals with faculty

Co-Founder, AIoT Tech

- Co-founded an AI services company delivering enterprise AI projects for clients including Bajaj and SKF

SELECTED INDEPENDENT PROJECTS

DeliverIQ — Delivery Intelligence System (9-Module AI/ML Suite)

Paper in preparation, IEEE Access

- Designed 9 interconnected AI/ML modules spanning route optimization, demand forecasting, anomaly detection, and CV-based package tracking
- Built an end-to-end RAG pipeline achieving 90%+ retrieval accuracy for delivery Q&A, with full benchmark evaluations

Enterprise RAG-Based Document Intelligence System

- Production RAG pipeline with recursive chunking, hybrid search, and metadata filtering, reducing document retrieval time by 70% with 90%+ LLM answer accuracy

Agentic AI — Multi-Agent Orchestration Framework

- Multi-agent system with tool-calling, persistent memory, and reflection loops deployed for 500+ daily sessions, cutting workflow completion time by 60% vs. a single-agent baseline

TECHNICAL SKILLS

- **Languages:** Python (Advanced), SQL, JavaScript
- **GenAI & LLMs:** LangChain, LangGraph, MCP, OpenAI API, Groq, Ollama, fine-tuning LLMs, prompt engineering, agentic AI
- **RAG & Vector Databases:** ChromaDB, FAISS, hybrid search, semantic embeddings, chunking & retrieval strategies
- **ML / DL:** PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers, OpenCV, deep reinforcement learning
- **Backend & Cloud:** FastAPI, Flask, Firebase, REST APIs, AWS, Docker, Git/GitHub, CI/CD
- **LLMOps:** evaluation pipelines, hallucination mitigation, guardrails, cost optimization, multi-agent orchestration

HONORS & CERTIFICATIONS

- TCS iON – Artificial Intelligence Pro (Certified)
- 60+ certifications across Deep Learning, GenAI, LLMOps, AWS, and Computer Vision (Coursera, DeepLearning.AI, LinkedIn Learning)
- 24+ end-to-end AI/ML project repositories on GitHub (CoderFatherBB)